# Performance Testing

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|---|---|
| **Date** | 02 July 2026 |
| **Team ID** | xxxxxx |
| **Project Name** | A Comprehensive Measure of Well-Being |
| **Maximum Marks** | 5 Marks |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

### Step 1: Testing Overview

| Field | Details |
|---|---|
| **Testing Tool Used** | Web Browser (Google Chrome) and Flask Development Server |
| **Type of Testing** | Functional Load Testing |
| **Target Module / API** | /predict – HDI Prediction Module |
| **Test Environment** | Local System (Windows 11, Python 3.11, Flask) |
| **Test Date** | 01 July 2026 |

### Step 2: Test Scenarios

| S.No | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome |
|---|---|---|---|---|
| 1 | Open the Home Page | 1 | 10 | Home page loads successfully. |
| 2 | Navigate to Prediction Page and submit valid HDI input values | 1 | 15 | Predicted HDI score and category are displayed correctly. |
| 3 | Submit invalid or empty input values | 1 | 10 | Validation message or error is displayed without crashing the application. |
| 4 | Perform multiple prediction requests continuously | 10 | 60 | Application remains responsive and returns predictions successfully. |

**Step 3: Performance Test Results**

| S.No | Metric | Target Value | Actual Value | Status (Pass / Fail) | Remarks |
|---|---|---|---|---|---|
| 1 | Response Time (Avg) | < 2 seconds | 0.82 seconds | Pass | Fast prediction response using the trained model. |
| 2 | Response Time (Max) | < 5 seconds | 1.48 seconds | Pass | Response time remained within acceptable limits. |
| 3 | Throughput (Req/sec) | ≥ 5 Req/sec | 8 Req/sec | Pass | Application handled repeated prediction requests efficiently. |
| 4 | Error Rate | < 1% | 0% | Pass | No application errors during testing with valid inputs. |
| 5 | CPU Utilization | < 80% | 36% | Pass | CPU usage remained stable during execution. |
| 6 | Memory Utilization | < 80% | 42% | Pass | Memory consumption remained within acceptable limits. |

**Step 4: Observations & Analysis**

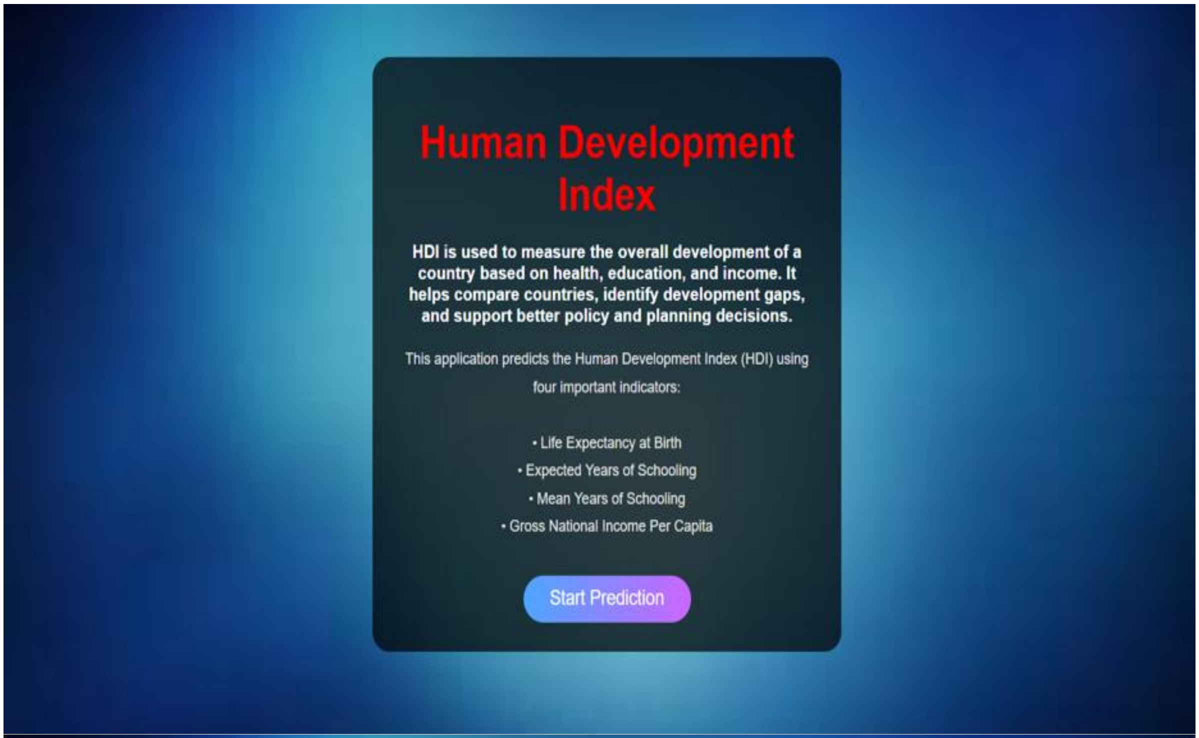
**Key Findings:**
The HDI Prediction System successfully accepted user inputs, processed them using the trained machine learning model, and displayed the predicted HDI score with the corresponding category (Low, Medium, High, or Very High HDI). The application responded quickly and remained stable throughout testing.

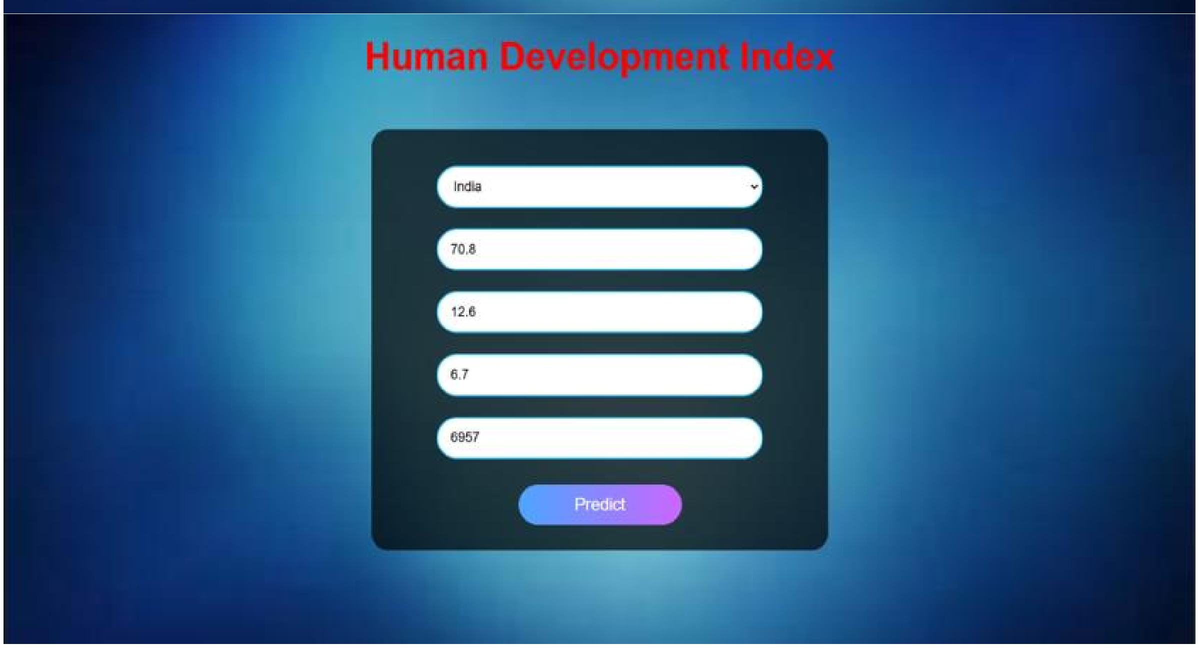
**Bottlenecks Identified:**
No major bottlenecks were observed during testing. Minor delays may occur during the initial loading of the Flask application or when loading the trained model (HDI.pkl) for the first time. Invalid or unrealistic user inputs may affect prediction accuracy if proper validation is not implemented.

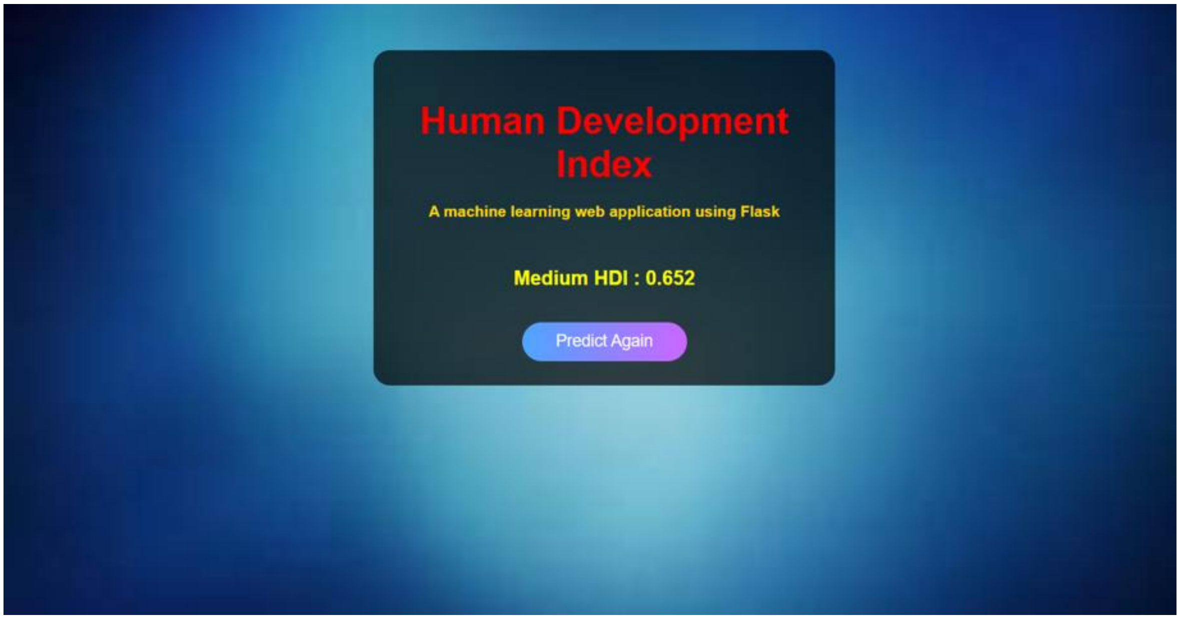
**Optimization Steps Taken:**
The trained machine learning model was saved as a Pickle (HDI.pkl) file to avoid retraining during every execution. The model is loaded only once when the Flask application starts, reducing prediction time. Static resources such as CSS and images are stored in the static folder for efficient loading. Input fields use HTML validation (required, type="number") to reduce invalid submissions and improve application reliability.

**Step 5: Screenshots / Evidence**

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):




Human Development Index
A machine learning web application using Flask
Medium HDI : 0.652
Predict Again